

Bias terms and decision boundaries

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Dot products and weighted sums

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Plot and label the boundary for Bias terms and decision boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Bias terms and decision boundaries. Proposed explanation: “Add bias to each contribution”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Analyse a new model independently

A new model has $w = [-1, 2]$, $b = -2$, inputs $[u, v]$ and class 1 when $z \geq 0$. Calculate scores and classes for $[2, 1]$, $[2, 2]$, $[2, 3]$. Write the boundary as v in terms of u . Then change only b to 0 and recompute $[2, 1]$.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace the bias once

Use $z = 2 \text{ times width} - \text{height} - 3$ at $[5, 4]$. Give the weighted sum, bias, final score and prediction under $z \geq 0$.

My reasoning and evidence

6. W2 Plot and label the boundary

For the worked model, find boundary points at width 2 and width 4. Test $[4, 4]$ and $[4, 6]$ to identify the class on each side.

My reasoning and evidence

7. W3 Check the equality rule

At $[3, 3]$, compare prediction under $z \geq 0$ with prediction under $z > 0$. Does changing this comparison move the zero line?

My reasoning and evidence

8. W4 Analyse a new model independently

A new model has $w = [-1, 2]$, $b = -2$, inputs $[u, v]$ and class 1 when $z \geq 0$. Calculate scores and classes for $[2, 1]$, $[2, 2]$, $[2, 3]$. Write the boundary as v in terms of u . Then change only b to 0 and recompute $[2, 1]$.

My reasoning and evidence

9. W5 Require evidence of improvement

A teammate says that raising b must improve the model because all scores rise. What information is needed before accepting an improvement claim?

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled bias terms and decision boundaries evidence sheet to Image-pattern Network Investigation.

Show how you can calculate a weighted score with a bias term. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a

changed case.

12. Pictorial case practice: Bias terms and decision boundaries

Move a toy boundary with a bias while keeping its input weights fixed.

Score $z=2u-v-3$; predict class 1 when $z \geq 0$. Inputs $A=[2, 1]$, $B=[1, 1]$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/3

Input A reaches the inclusive boundary

Input 2 x weight 2 = 4

Input 1 x weight -1 = -1

Add the products and bias -3: total = 0

Keep the total (identity) -> output 0

Bias terms and decision boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Trace input A

A score= $4-1-3=0$, so class 1 at the declared inclusive boundary.



2. Trace input B

B score= $2-1-3=-2$, so class 0.



3. Read boundary

The boundary is $v=2u-3$; the bias sets its position.

CHANGE ONE THING

Change bias from -3 to -1.

Explain the changed result using the matching lesson.

Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Move a toy boundary with a bias while keeping its input weights fixed.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The boundary is $v=2u-3$; the bias sets its position.

1. Trace input A
2. Read boundary
3. Trace input B

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Read boundary The boundary is $v=2u-3$; the bias sets its position. The bias adds the same offset after the input contributions. It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule. That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary.

Practice 2: explain the deciding evidence

Changed-case practice: Change bias from -3 to -1. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. A score becomes 2 and B score 0; both predict class 1.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A score becomes 2 and B score 0; both predict class 1. Increasing bias by 2 adds 2 to every score without changing the weights.

Repair a missing building block

Review Dot products and weighted sums ($a_i \cdot 5/2$). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
